Shamitha R

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Professional Summary

ECE graduate with hands-on experience in VLSI, RTL design, Verilog HDL, digital logic, functional verification, and embedded systems. Skilled in developing FSM-based digital designs, RTL simulation, waveform analysis, and microcontroller-based hardware interfacing. Familiar with EDA and simulation tools including Xilinx Vivado, LTspice, Cadence, and Synopsys. Seeking entry-level opportunities in RTL Design, VLSI Verification, Digital Design, or Embedded Systems.

Education

Bachelor of Engineering in Electronics and Communication Engineering **CGPA: 8.29**
KSR Institute for Engineering and Technology, Tiruchengode

Technical Skills

- **HDL & Programming:** Verilog HDL, VHDL, Embedded C, C
- **RTL & Digital Design:** RTL Design, FSM, Digital Logic, Combinational and Sequential Circuits
- **EDA & Simulation Tools:** Xilinx Vivado, LTspice, Cadence, Synopsys
- **Embedded Systems:** ESP32, ESP8266, Microcontrollers, Sensor Interfacing, Peripheral Interfacing, Arduino IDE

VLSI / RTL Projects

Traffic Light Control System – Verilog HDL

- Designed an FSM-based traffic light controller using Verilog HDL for automated traffic signal sequencing.
- Developed RTL logic with defined states and transition conditions for controlling traffic light outputs.
- Implemented combinational and sequential logic to manage state transitions and signal timing.
- Simulated the RTL design using EDA tools and analyzed waveforms to verify functional correctness.

Embedded / IoT Projects

Smart Fire and Gas Detection System – ESP32

- Developed an ESP32-based IoT system for real-time detection of gas leakage and flame conditions.
- Implemented threshold-based firmware logic to identify hazardous conditions and initiate alert mechanisms.
- Integrated sensors with the microcontroller for continuous environmental monitoring and real-time response.
- Tested system behavior under different sensor conditions to improve reliability of hazard detection.

Smart Lock System – ESP8266

- Developed an ESP8266-based smart lock system for secure and controlled access.
- Programmed the microcontroller using Arduino IDE to process inputs and control system outputs.
- Interfaced input and output components to achieve real-time access control and system response.
- Tested hardware functionality and verified reliable operation under different input conditions.

Internship Experience

VLSI Intern – Caliber Technology, Coimbatore

- Designed and simulated digital circuits using Verilog HDL as part of VLSI design activities.
- Developed RTL logic and performed functional verification through simulation.
- Gained practical exposure to RTL design, digital circuit simulation, and VLSI development workflows.

IoT Intern – Samcore Solutions, Trichy

- Developed and tested microcontroller-based IoT systems for real-time monitoring applications.
- Programmed embedded systems using Embedded C and performed hardware and sensor interfacing.
- Worked with input and output peripherals for real-time system response.
- Gained practical exposure to embedded system development, communication, and hardware integration.

Certifications

- **NPTEL** – Internet of Things
- **Functional Verification** – SystemVerilog and UVM
- **RTL Design** – Verilog HDL

Areas of Interest

VLSI Design, RTL Design, Functional Verification, Digital Design, Embedded Systems